Meven Lennon-Bertrand

Maître de conférences – ENS Rennes

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Logic | Dependent types | Proof assistants | Typing algorithms | Formalised mathematics

Selected Research Highlights

Journal Articles

- Sozeau, Forster, Lennon-Bertrand et al., *Correct and Complete Type Checking and Certified Erasure for Coq, in Coq* (JACM 2025)
- Lennon-Bertrand, Maillard, Tabareau et al., *Gradualizing the Calculus of Inductive Constructions* (TOPLAS 2022)

Conference Articles

- Adjedj, Lennon-Bertrand, Benjamin et al., *AdapTT: Functoriality for Dependent Type Casts* (POPL 2026)
- Lennon-Bertrand, *What Does It Take to Certify a Conversion Checker?* (FSCD 2025, **Best paper by junior researcher**)
- Adjedj, Lennon-Bertrand, Maillard et al., *Martin-Löf à la Coq* (CPP 2024, **Distinguished paper**)

Formalisations

MetaRocq | LogRel-Rocq | Coq-PartialFun | AutoSubst 2

Invited Talks

Workshop on the Implementation of Type Systems
Jan. 2026 — Rennes

Meeting of the EuroProofNet Working Group 6
Apr. 2024 — KU Leuven

Student Supervision

Simon Corbard – Coercions in Dependent Type Theory
ENS Paris-Saclay M2 Internship, then PhD thesis
Mar. 2026 – … — Co-supervised with Thibaut Benjamin

Arthur Adjedj – Subtyping in Dependent Type Theory
ENS Paris-Saclay long research internship
Oct. 2024 – Aug. 2025

Community Service

Program Committee

ICFP '27 | RocqShop '26 | JFLA '26 | ITP '25 | Types '24

Types Steering Committee
2025 – 2028

Positions

Maître de Conférences
2026 – … — ENS Rennes

Inria Starting Researcher Position
2025 – 2026 — IRIF, Université Paris Cité

Research Associate
2022 – 2025 — University of Cambridge

PhD Student
Supervisor: Nicolas Tabareau
2019 – 2022 — Inria, Université de Nantes

Selected Teaching & Outreach

Formal proof and synthetic maths workshop
Jun. 2026 — Heidelberg University

Lecturer: Proof Assistants
2024 — University of Cambridge

Lecturer: Denotational Semantics
2023, 2024 — University of Cambridge

Science Popularisation: CHantiers Arts, Sciences et Technologies
2019 – 2022 — Lycée Michelet, Nantes

Education

Master 2 (Computer Science)
2018 – 2019 — ENS de Lyon

Master 2 (Mathematics)
Preparation to the Agrégation, received 10th
2017 – 2018 — ENS de Lyon

Master 1 (Math. Foundations of Comp. Sci.)
2016 – 2017 — Nijmegen, Erasmus exchange

Bachelors (Computer Science & Maths)
2015 – 2016 — ENS de Lyon

Publications

I generally use the first author position to highlight key authors. To distinguish when this is the case (even when authors seem to be alphabetically sorted), I use an asterisk in the list below.

Journal

Soz+25 Correct and Complete Type Checking and Certified Erasure for Coq, in Coq
2025 Matthieu Sozeau*, Yannick Forster, Meven Lennon-Bertrand, Jakob Nielsen, Nicolas Tabareau and Théo Winterhalter. *Journal of the ACM*. DOI: 10.1145/3706056.

Len+22 Gradualizing the Calculus of Inductive Constructions
2022 Meven Lennon-Bertrand*, Kenji Maillard*, Nicolas Tabareau and Éric Tanter. *ACM Transactions on Programming Languages and Systems*. DOI: 10.1145/3495528.

Conference post-proceedings

LS26 Bidirectional Interpolation for the λ-Calculus – Revisiting and Formalising Craig-Čubrić Interpolation
2026 Meven Lennon-Bertrand and Alexis Saurin. *17th International Conference on Interactive Theorem Proving*. DOI: 10.4230/LIPIcs.ITP.2026.30.

Adj+26 AdapTT: Functoriality for Dependent Type Casts
2026 Arthur Adjedj*, Meven Lennon-Bertrand*, Thibaut Benjamin and Kenji Maillard. *Proceedings of the ACM on Programming Languages*. DOI: 10.1145/3776664.

SLK25 Implementing a Type Theory with Observational Equality, Using Normalisation by Evaluation
2025 Matthew Sirman*, Meven Lennon-Bertrand and Neel Krishnaswami. *Post-Proceedings of the 30th International Conference on Types for Proofs and Programs*. DOI: 10.4230/LIPICS.TYPES.2024.5.

Len25 What Does It Take to Certify a Conversion Checker?
2025 Meven Lennon-Bertrand. *10th International Conference on Formal Structures for Computation and Deduction*. **Best paper by junior researcher**. DOI: 10.4230/LIPIcs.FSCD.2025.27.

LLM24 Definitional Functoriality for Dependent (Sub)Types
2024 Théo Laurent*, Meven Lennon-Bertrand and Kenji Maillard. *33rd European Symposium on Programming*. **Distinguished artefact**. DOI: 10.1007/978-3-031-57262-3_13.

Adj+24 Martin-Löf *à la* Coq
2024 Arthur Adjedj, Meven Lennon-Bertrand, Kenji Maillard, Pierre-Marie Pédrot and Loïc Pujet. *Proceedings of the 13th ACM SIGPLAN International Conference on Certified Programs and Proofs*. **Distinguished paper**. DOI: 10.1145/3636501.3636951.

Mai+22 A Reasonably Gradual Type Theory
2022 Kenji Maillard*, Meven Lennon-Bertrand, Nicolas Tabareau and Éric Tanter. *International Conference on Functional Programming*. DOI: 10.1145/3547655.

Len21 Complete Bidirectional Typing for the Calculus of Inductive Constructions
2021 Meven Lennon-Bertrand. *12th International Conference on Interactive Theorem Proving*. DOI: 10.4230/LIPIcs.ITP.2021.24.

PhD Thesis

Len22 Bidirectional Typing for the Calculus of Inductive Constructions
2022 Meven Lennon-Bertrand.

Peer-reviewed workshops

Len26 Verifying Dependent Type-Checkers
2026 Meven Lennon-Bertrand. *5th Workshop on the Implementation of Type Systems.* **Invited talk**. URL: https://popl26.sigplan.org/home/wits-2026.

Adj+25 AdapTT: A Type Theory with Functorial Types
2025 Arthur Adjedj*, Meven Lennon-Bertrand*, Thibaut Benjamin and Kenji Maillard. *31st International Conference on Types for Proofs and Programs.* URL: https://msp.cis.strath.ac.uk/types2025/.

Len24 Towards a certified proof assistant kernel
2024 Meven Lennon-Bertrand. *Meeting of the Working Group 6 of the European Research Network on Formal Proofs.* **Invited talk**. URL: https://europroofnet.github.io/wg6-leuven/.

SLK24 Implementing Observational Equality Using Normalisation by Evaluation
2024 Matthew Sirman*, Meven Lennon-Bertrand and Neel Krishnaswami. *30th International Conference on Types for Proofs and Programs.* URL: https://types2024.itu.dk/Index.html.

LK23 Decidable Type-Checking for Bidirectional Martin-Löf Type Theory
2023 Meven Lennon-Bertrand* and Neel Krishnaswami. *29th International Conference on Types for Proofs and Programs.* URL: https://types2023.webs.upv.es/.

Mai+23 Engineering logical relations for MLTT in Coq
2023 Kenji Maillard, Arthur Adjedj, Meven Lennon-Bertrand and Loïc Pujet. *29th International Conference on Types for Proofs and Programs.* URL: https://types2023.webs.upv.es/.

Len22a Equivalence between Typed and Untyped Algorithmic Conversions
2022 Meven Lennon-Bertrand. *28th International Conference on Types for Proofs and Programs.* URL: https://types22.inria.fr/programme/.

Len22b À bas l'η – Coq's troublesome η-conversion
2022 Meven Lennon-Bertrand. *1st Workshop on the Implementation of Type Systems.* URL: https://popl22.sigplan.org/home/wits-2022#event-overview.

SLF22 The Curious Case of Case: Correct and Efficient Representation of Case Analysis in Coq and MetaCoq
2022 Matthieu Sozeau, Meven Lennon-Bertrand and Yannick Forster. *1st Workshop on the Implementation of Type Systems.*

Publication in open archives

BR18 Coalgebraic Determinization of Alternating Automata
2018 Meven Bertrand and Jurriaan Rot. DOI: 10.48550/ARXIV.1804.02546.

Software Development: Formalisation Projects

Contributions are described using Inria's criteria for software evaluation. All my software is open access.

MetaRocq

Family = research; Audience = community; Evolution = lts; Duration ≥ 5; Contribution = devel, softcont

2020–… The MetaRocq team

Large collaborative project aiming at formalizing Rocq in Rocq itself, and to allow manipulating Rocq terms in Rocq in order to develop certified meta-programming tools.

LogRel-Rocq

Family = research; Audience = partners; Evolution = lts; Duration ≥ 3; Contribution = leader, devel, softcont

2022–… A. Adjedj, K. Maillard, P.-M. Pédrot, L. Pujet

Formalisation of a proof of normalisation by logical relations and verification of a type checker, for a dependent type theory.

AutoSubst 2

Family = utility; Audience = partners; Evolution = basic; Duration ≥ 3; Contribution = softcont

2023–… Adrian Dapprich, Yannick Forster, Kathrin Stark

Code generator dedicated to syntax with binders.

Coq-PartialFun

Family = utility; Audience = partners; Evolution = basic; Duration ≥ 3; Contribution = softcont

2023-… Théo Winterhalter, Kenji Maillard

Support library for the definition of non-structurally recursive functions.

Article artefacts

2022–…

Many conferences in the programming language community organise *Artefact evaluation committee*, which evaluate artefacts (software, formalisations, etc.) independently of program committees. Most of my papers have had their accompanying code reviewed by such artefact evaluation committees, and have systematically been awarded the best available level of evaluation.

- Artefact for 'Definitional Functoriality for Dependent (Sub)Types' [LLM24]:
  Family = research; Audience = partners; evolution = nofuture; Duration = 1; contribution = leader; Url = https://zenodo.org/records/10508084
- Artefact for 'Martin-Löf *à la* Coq' [Adj+24]:
  Family = research; Audience = partners; evolution = nofuture; Duration = 1; contribution = leader; Url = https://zenodo.org/records/8367154
- Artefact for 'A Reasonably Gradual Type Theory' [Mai+22]:
  Family = research; Audience = partners; evolution = nofuture; Duration = 1; contribution = devel; Url = https://zenodo.org/records/6928465
- Artefact for 'Gradualizing the Calculus of Inductive Constructions' [Len+22]:
  Family = research; Audience = partners; evolution = nofuture; Duration = 1; contribution = devel; Url = https://github.com/pleiad/GradualizingCIC

Supervised students

Main supervisor

Simon Corbard – Coercions in Dependent Type Theory

PhD thesis

Sep. 2026 – … Co-supervised with Thibaut Benjamin and Sam van Gool

Simon Corbard – Meta-theory of a Type Theory with Adapters
M2 Internship
Mar. 2026 – Jul. 2026 Co-supervised with Thibaut Benjamin

Arthur Adjedj – Subtyping in Dependent Type Theory
ENS Paris-Saclay long research internship
Oct. 2024 – Aug. 2025

Secondary supervisor

Adrien Mathieu – Domain Models of Dependent Type Theory
ENS Paris long research internship
Oct. 2025 – Jul. 2026 Co-supervised with Paul-André Melliès

Ilya Kaysin – Complete Unification for Dependent Type Inference
PhD Student
Oct. 2024 – … Main supervisor: Neel Krishnaswami

Robin Jourde – Understanding the η Law for Functions in CIC
Master 2 internship
Jan. – Jul. 2023 Co-supervised with Nicolas Tabareau

Matthew Sirman – Normalisation by Evaluation for Observational Equality
Undergraduate final dissertation (4^{th} year)
Nov. 2022 – May 2023 Co-supervised with Neel Krishnaswami

Talks and Seminars

Invited Talks

Formal structures to represent proofs and programs
Joint King's College London and Université Paris Cité workshop
Feb. 2026 Université Paris Cité

Workshop on the Implementation of Type Theory
Jan. 2026 Rennes

Formalisation of Mathematics with Interactive Theorem Provers
Nov. 2024 Faculty of Mathematics, Cambridge

Big Specification Workshop
Oct. 2024 Newton Institute, Cambridge

ReCiProg ANR Workshop Day
Jun. 2024 — Université de Nantes

EuroProofNet Working Group 6 Meeting – Invited Talk
Apr. 2024 — KU Leuven

CHoCoLa Seminar
Jan. 2023 — ENS de Lyon

Research Visits

Thierry Coquand & Logic and Types group
Apr. – May 2026 — Chalmers University of Technology

Kathrin Stark & Dependable Systems Group
Mar. 2024 — Heriot-Watt University

Conor McBride & Mathematically Structured Programming Group
Jun. 2023 — University of Strathclyde

Andrej Bauer & Faculty of Mathematics and Physics Foundations Seminar
May 2022 — University of Ljubljana

Invited Team/Lab Seminars

Antique, ENS Paris · LIFO, Université d'Orléans · CASH, ENS de Lyon · Épicure, IRISA, Rennes · OASIS, University of Oxford · Deducteam, LMF, Paris-Saclay · LIX, École Polytechnique · PPS, IRIF, Université Paris Cité · Formath, Inria Paris · LoVE, LIPN, Université Sorbonne Paris Nord

Academic and community service

Reviewing Organisation

Program Committee
ICFP 2027 · RocqShop 2026 · JFLA 2026 · ITP 2025 · Types 2024

Journal Reviewer
LMCS

Conference Subreviewer
CSL 2026 · FSCD 2026 · CPP 2026 · POPL 2026 · LICS 2025 · FSCD 2025 · CPP 2025 · CSL 2025

Artefact Evaluation Committee

ICFP 2022

Community Organisation

Types Steering Committee

2025 – 2028

In charge of the long-term organisation and supervision of the Types conference. I was elected a member during the general assembly of the 2025 edition, and will serve for 3 years.

Formath Seminar (organiser)

2026 – 2027 IRIF, Paris

Seminar of the Picube team.

SANDWICH Seminar (organiser)

2024 – 2025 University of Cambridge

Internal seminar of the CLASH group.

Proof Assistants Stack Exchange

2022 – ...

This website aims to answer questions around proof assistants in a community-based manner. I am in the top 5 most reputable users, and a moderator.

Representation

Post-doc Representative – Lab Council

2026 – 2027 IRIF, Paris

Elected Student Representative

2017 – 2018 ENS de Lyon

Teaching and Outreach

Undergraduate and Graduate Level

Co-Lecturer: Proof Assistants

Fall 2024 University of Cambridge

With Thomas Bauereiss, for 4th year students. I was in charge of the Rocq half of the course.

Lecturer: Denotational Semantics

Fall 2023, Fall 2024 University of Cambridge

For two years in a row, I lectured the Denotational Semantics course for 3rd year students.

Teaching Assistant (64 hrs/year)

2019 – 2022 Université de Nantes

I taught various levels (1st to 3rd Bachelor years), topics (mathematics, applied and fundamental computer science), formats (lectures, exercise and computer sessions), and audiences (specialists and non-specialists).

Co-Lecturer: Category Theory

Sept 2018 – Jan 2019 ENS de Lyon

During my Master 2, I co-lectured a category theory course for ENS de Lyon students.

Research Level

Formal proof and synthetic mathematics workshop

Jun. 2026 Heidelberg University

A workshop to introduce type theory and synthetic mathematics to an audience of (mainly) mathematics researchers and students.

Outreach

CHantiers Arts, Sciences et Technologies

2019 – 2022 Théâtre Athénor & Lycée Professionnel Michelet, Nantes

Together with mathematician Bertrand Michel and writers Rémi Checchetto and Sylvain Renard, we collaborated with a vocational high-school teachers to build workshops for their students. Sylvain and I have been writing on this experience since, hopefully it will be eventually published...

Séminaire de la Détente Mathématique

2018 – 2019 Maison des Mathématiques et de l'Informatique, Lyon

A weekly seminar, aimed at being relaxed and accessible to both students and faculty, with talks often on unusual or fun topics. Many students would give their first seminar talk ever there. I organised the seminar with a team of students, and spoke there.

Thesis

Defended at Université de Nantes, on June 24, 2022. Prepared in the Inria team Gallinette, affiliated to the Laboratoire des Sciences du Numérique de Nantes.

PhD Advisor	Nicolas Tabareau	Directeur de Recherche, Inria Rennes Bretagne Atlantique
Head of the jury	Christine Paulin-Mohring	Professeure des Universités, Université Paris Sud
Rapporteurs	Neel Krishnaswami	Associate Professor, University of Cambridge
	Conor McBride	Reader, University of Strathclyde
Examiners	Jesper Cockx	Assistant Professor, TU Delft
	Herman Geuvers	Professor, Radboud University Nijmegen
	Hugo Herbelin	Directeur de Recherche, Inria Paris
	Assia Mahboubi	Directrice de Recherche, Inria Rennes Bretagne Atlantique
Invited Member	Matthieu Sozeau	Chargé de Recherche, Inria Rennes Bretagne Atlantique